

Improving the Student Account Password and Multifactor Authentication Process

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To: Director of Information Technology

From: Latasha Kellon

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Universities have a duty to protect student accounts because these accounts often contain access to email, coursework, financial and academic information, and other sensitive resources. The instructional handout created in Assignment 1 described how users can create a strong password and enable multifactor authentication (MFA). This process offers a workable baseline for account security; however, several alternatives would make the authentication process both more secure and more usable for students. This report assesses the usability of the current process and compares it to stronger authentication alternatives. This report also offers recommendations for improvements the university's information technology administrators can make.

Evaluation of the Current Process

There are multiple things that make the current process strong. Students are given step-by-step instructions on how to make a password and enable MFA. By giving users individual steps to follow, the process decreases the amount of information a user must simultaneously process. The last way this process is effective is that it adds protection beyond a singular password. According to the Cybersecurity and Infrastructure Security Agency (CISA), MFA adds further security because "if a password is compromised, it alone does not provide an attacker with the additional authentication factor necessary to access the account" (CISA, n.d.).

It should be noted that the process can be improved on. Users, in this case students, are still able to make predictable passwords or the same password for several accounts. It can also be an aggravating process if the guidelines do not sufficiently instruct how to recover accounts, what backup authentication processes are, or what to do after losing the device with which the account is registered. The existing NIST guidelines require single factor authentication passwords to be at least 15 characters long and not have arbitrary composition rules. The NIST also recommends password managers as they can generate and store different unique passwords for sites (National Institute of Standards and Technology [NIST], 2025a, 2025b).

Comparison of Authentication Approaches

The current password-and-MFA model is significantly more robust than passwords alone. However, all MFA is not created equal. The stronger forms of MFA provide a level of phishing resistance. CISA has noted security keys are stronger than text or email codes and have a goal of organizations moving to phishing-resistant MFA (CISA, n.d.). NIST also indicates that passwords alone are not phishing-resistant and that cryptographic methods of authentication can be phishing resistant (NIST, 2025a).

Table 1

Comparison of Student Authentication Approaches

Approach	Security	Convenience	Primary Limitation
Password only	Low	High	Password may be stolen, guessed, or reused
Password + SMS MFA	Moderate	High	SMS provides weaker protection
Password + authenticator app	High	Moderate	Requires access to registered device

Passkey/security key	Very High	High after setup	Initial enrollment may require guidance
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Recommendations

The university should keep MFA in place and work towards improving the enrollment process over time. Students should be nudged to use password managers and long, unique passwords instead of hard-to-remember password patterns. The university should also enable students to use authenticator apps or phishing-resistant methods like passkeys or security keys as soon as its systems allow these options.

Instructions need to show screenshots, clear steps with numbers, troubleshooting tips, and account-recovery processes. I think students should also be provided a brief explanation as to why the security measure is in place. I could see this increasing compliance, as the user is now aware of why an extra authentication process is in place rather than thinking of them as hurdles. In the event that an organization cannot immediately roll out phishing-resistant authentication, I see stronger app-based MFA as a temporary solution (CISA, n.d.).

Conclusion

The current password-and-MFA process is a good starting point to protecting student accounts. However, using better authentication technology and clearer user instructions would make it even stronger. Long unique passwords (entered or automatically filled by password managers), MFA, phishing-resistant authentication, and easy-to-understand recovery instructions could provide additional security without needlessly burdening students. The recommendations above would help the university balance security with usability in its authentication process.

Rationale

Assignments 1 and 2 both informed the writing of this report. Assignment 1 involved writing about password creation and multifactor authentication from the perspective of a user, and I realized that this missing contextual information when I was writing it. Assignment 2 enabled me to compare my user perspective to professional cybersecurity instruction and to see how I could improve on my work. They both showed me that good technical communication, especially instructions, required more than simply good information. Usability, audience, organization, and familiarity with current professional practices were also necessary. I used these insights to inform my recommendations on authentication that weigh student convenience and cybersecurity.

References

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